

Text Judges Can’t Hear: Prosodic Contrast Sets Expose Systematic Bias in Voice AI Evaluation

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Abstract. Text-based LLM judges have become a standard evaluation mechanism for voice AI, yet transcript-only evaluation cannot directly verify evidence carried by the waveform. We introduce Prosodic Contrast Sets (PCS), lexically matched audio variants whose pragmatic meaning changes with prosody, together with differentiation and direction-sensitive pragmatic accuracy. We expand PCS from 124 to 216 English items across 12 categories and evaluate Qwen3-Omni, GPT-4o-audio, Gemini Live, text-only systems, and ASR cascades with Qwen, Llama, and Mistral judges. On the expanded set, native differentiation is 0.664 versus 0.448 for text-only, a gap of 0.216 (95% CI [0.184, 0.248], Holm-adjusted $p < 0.001$, $d_z=1.11$). Across model families, audio-native systems average 0.641, compared with 0.456 for text-only and 0.471 for cascades. Pragmatic-label accuracy is 78.9% for native voice, 43.7% for text-only, and 46.1% for the cascade. The effect persists with two non-Qwen TTS engines, human speech, and a 30-item Mandarin pilot. A five-rater study over 60 items prefers audio-native responses in 271 of 296 non-tie judgments on marked variants (91.6%), and differentiation correlates with human appropriateness ($\rho=0.72$). Giving the judge audio raises native appropriateness from 39.6% to 81.3%. These results establish modality bias across sources, model families, prompts, and judges, motivating audio-aware evaluation for voice systems.

Keywords: Voice AI evaluation · Prosody · LLM-as-judge · Modality bias · Benchmark methodology.

1 Introduction

A 35-billion-parameter text judge scores a native voice model at **3.6%** accuracy, while a 9B judge gives the same model **29.8%**. Yet the expanded evaluation shows native differentiation of 0.664 versus 0.448 for text-only, and the pattern replicates across non-Qwen systems and judges.

This paradox emerges from a structural property of contemporary voice AI evaluation: text-based judges, now widely used across automated evaluation pipelines [6], are architecturally incapable of perceiving the prosodic information that distinguishes competent voice understanding from mere transcription. When a user says *“That’s fine”*, sincerely versus sarcastically, the appropriate conversational response differs dramatically. The distinguishing signal lies entirely in prosody: pitch contour, speaking rate, and energy distribution. Text

judges receive only the transcript and at most a brief textual prosody label, yet they evaluate systems whose responses depend on the full audio waveform.

The consequences are systematic. Modality bias intensifies with judge scale (accuracy drops from 29.8% at 9B to 3.6% at 35B), applies uniformly across twelve prosodic categories, and actively penalizes the very capability under evaluation. Recent voice and audio benchmarks increasingly test speech interaction, reasoning, and agentic behavior [12,30,15,10,19], but still rarely isolate response appropriateness under ASR-equivalent prosodic minimal pairs, meaning their rankings can systematically favor systems that ignore prosodic information.

We frame this as an *evaluation methodology problem*, not a model deficiency. We make four contributions. (i) **PCS**: a controlled methodology for testing prosodic responsiveness while holding lexical content fixed. (ii) **Complementary metrics**: differentiation, pragmatic accuracy, and repeated-stimulus consistency distinguish sensitivity from appropriate interpretation. (iii) **Cross-family validation**: three audio-native systems, non-Qwen judges, multiple TTS engines, human speech, and a Mandarin pilot. (iv) **Human and judge-modality validation**: a five-rater study and a direct comparison showing that audio access raises native appropriateness from 39.6% to 81.3%.

2 Related Work

Voice AI benchmarks. Recent benchmarks have broadened the evaluation surface for speech and audio systems: VocalBench [12] (conversational abilities), SpeechR [30] (speech reasoning with stress/emotion conditions), MMAR [15] (deep reasoning over speech, sound, and music), SSEU-Bench [33] (speech/scene/event understanding), and VoiceAgentBench [10] and τ -Voice [19] (agentic and full-duplex settings). These make important progress beyond ASR-only evaluation, but none isolate prosodic discrimination as ASR-equivalent lexical minimal pairs where the appropriate response depends on *how* something is said rather than *what* is said. Without PCS-style items, a system that merely transcribes can score identically (or higher, as our results show) to one that perceives vocal intent.

LLM-as-judge bias. Recent surveys and perturbation studies show that LLM-based evaluation remains vulnerable to position, verbosity, self/family, authority, and bandwagon biases [6,31,24,22]. We identify an orthogonal axis, modality bias, in which a judge must assess behavior grounded in evidence it cannot observe. Text captions and structured acoustic metadata partially reduce this mismatch, while audio-aware judging addresses it directly. Our claim is therefore not that all text-assisted evaluation inevitably fails, but that transcript-only judges cannot independently verify waveform-level prosodic evidence.

Prosody and minimal pairs. Our methodological foundation is the minimal pair tradition in phonology [16]. BLiMP [25] extended minimal pairs to syntactic competence; we extend further to prosodic competence. Prosody has been

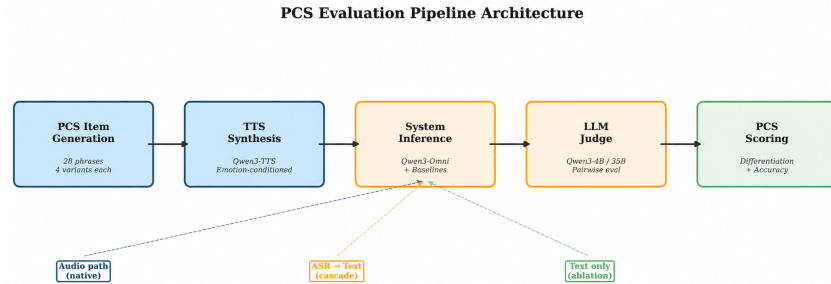


Fig. 1: Overview of the PCS evaluation framework. The box counts and Qwen-only labels depict the original pilot used to establish the pipeline; all camera-ready results use the expanded 216-item, cross-family evaluation described in Sections 4–5.

studied in speech emotion recognition [21], and its communicative role is well established [7,3]. Recent omni-modal systems increasingly process and generate speech end-to-end [4,27,13,28]. Closely related to our motivation, LISTEN disentangles lexical and acoustic emotion cues and finds that current audio LLMs often rely heavily on lexical semantics [1]. Our contribution focuses on whether systems appropriately *respond* to prosody rather than merely classify or detect it.

3 Theory and Method

3.1 Prosodic Contrast Sets

Definition 1 (Prosodic Contrast Set). A *Prosodic Contrast Set* $\mathcal{P} = (a_1, a_2, \dots, a_k)$ is a tuple of audio stimuli satisfying: (i) all a_i share identical lexical content w , i.e., $\text{ASR}(a_i) = w$ for all i ; (ii) stimuli differ in at least one prosodic dimension $\delta \in \{\text{pitch}, \text{speed}, \text{energy}\}$; (iii) the prosodic difference induces a pragmatically distinct interpretation requiring a distinct appropriate conversational response.

By construction, any observed difference in system behavior across PCS variants *must* be attributable to prosodic perception, since all other variables are held constant. We construct PCS items via a three-stage pipeline (Algorithm 1), illustrated in Fig. 1.

Stage 1: Linguistic seed curation. We identify utterances whose pragmatic interpretation is prosody-dependent, that is, sentences ambiguous in written form but unambiguous when spoken with specific prosody. A valid seed admits at least two pragmatically distinct interpretations that map to distinct appropriate responses.

Stage 2: Prosodic contrast specification. For each seed, we define target prosodic realizations along pitch contour (rising, falling, flat), speaking

Algorithm 1 PCS Construction Pipeline**Require:** Linguistic seeds S , TTS system T , prosodic specifications P **Ensure:** Validated PCS dataset D

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1: for each seed  $s \in S$  do
2:   Define  $k$  prosodic contrasts  $C = \{c_1, \dots, c_k\}$  from  $P$ 
3:   for each contrast  $c_i \in C$  do
4:     Synthesize:  $a_i \leftarrow T(s, c_i)$ 
5:     Verify prosodic realization (acoustic analysis)
6:   end for
7:   Validate:  $\text{ASR}(a_i) = \text{ASR}(a_j) \forall i, j$ 
8:   Add  $(a_1, \dots, a_k)$  to  $D$ 
9: end for
10: return  $D$ 

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rate (fast/clipped vs. slow/deliberate), and energy distribution (emphatic stress, monotone, trailing off). Each configuration is paired with its expected pragmatic interpretation.

Stage 3: Variant synthesis and validation. The main set uses Qwen3-TTS [9]; replication subsets use two non-Qwen TTS engines and human speech from multiple speakers and speaking styles. Acoustic verification confirms the intended pitch, energy, and rate contrasts, while Whisper large-v3 [18] verifies semantic ASR equivalence. Human listeners recover intended interpretations of synthesized variants with 86.4% accuracy (Fleiss’ $\kappa=0.71$).

The final dataset comprises **216 PCS items**, with at least 15 items in each of 12 pragmatic categories: sarcasm, enthusiasm, hesitation, urgency, boredom, anger, disappointment, surprise, confidence, doubt, affection, and indifference. A separate 30-item Mandarin pilot tests cross-lingual transfer.

3.2 Differentiation Metric

Definition 2 (Differentiation). For a system S and PCS item $\mathcal{P} = (a_1, \dots, a_k)$, let $r_i = S(a_i)$. The differentiation score is

$$\mathcal{D}(S, \mathcal{P}) = \frac{2}{k(k-1)} \sum_{i < j} (1 - \cos(\mathbf{e}(r_i), \mathbf{e}(r_j))), \quad (1)$$

where $\mathbf{e}(\cdot)$ is a sentence embedding and $\cos(\cdot, \cdot)$ is cosine similarity.

System-level differentiation is the mean across all PCS items:

$$\mathcal{D}(S) = \frac{1}{|D|} \sum_{\mathcal{P} \in D} \mathcal{D}(S, \mathcal{P}). \quad (2)$$

Definition 3 (Discrimination Criterion). A system S demonstrates genuine prosodic discrimination iff: (i) $\mathcal{D}(S) > \mathcal{D}(\text{text-only})$ at statistical significance; and (ii) self-similarity under repeated identical stimuli $\rightarrow 1$, indicating deterministic response to a given prosodic input.

Table 1: Expanded PCS results. Differentiation measures response sensitivity; pragmatic accuracy tests whether the response changes in the intended direction.

System	Diff.	Pragmatic acc.
Qwen3-Omni (native)	0.664	78.9%
ASR cascade	0.471	46.1%
Text-only	0.448	43.7%

The native system combines high differentiation (0.664), 78.9% pragmatic accuracy, and stable repeated responses. A random-response baseline reaches high differentiation (0.741) but only 18.6% pragmatic accuracy and fails consistency. Differentiation must therefore be paired with direction-sensitive validity rather than treated as correctness by itself.

4 Experimental Setup

Dataset. The expanded English benchmark contains 216 PCS items across 12 categories, with at least 15 items per category. Source-replication subsets use two non-Qwen TTS engines and human speech; a 30-item Mandarin pilot tests cross-lingual transfer.

Systems. Audio-native evaluation includes Qwen3-Omni [28], GPT-4o-audio, and Gemini Live. Text-only systems receive the transcript; cascades use Whisper large-v3 ASR before response generation; Text+Caption receives the transcript and explicit prosody description. Response generation uses temperature 0.7, with a temperature-0 diagnostic to isolate sampling effects.

Judges. Transcript-only judges include Qwen3-9B and Qwen3-35B, Llama-3.3-70B, and Mistral-Large. We evaluate three semantically equivalent prompts. The audio-aware judge receives the original waveform and is disjoint from the evaluated response systems. In Text+Caption, the judge sees transcript and caption but not waveform audio.

Human evaluation and robustness. Five independent raters evaluate 60 PCS items spanning all 12 categories. We report preference, pragmatic-label accuracy, inter-rater agreement, and association with differentiation. Robustness checks include MiniLM, MPNet, and BGE embeddings, length normalization, paraphrase/style controls, three prompts, multiple decoding settings, non-Qwen synthesis, human speech, and Mandarin. Paired-samples effect sizes are reported as Cohen’s d_z .

5 Main Results

5.1 Differentiation Hierarchy and the Text-Bias Paradox

The expanded benchmark preserves the original effect (Table 1): native voice reaches 0.664 differentiation versus 0.448 for text-only, a gap of 0.216 (95%



Fig. 2: The text-bias paradox. (a) Differentiation across systems with confidence intervals. (b) The original Qwen 9B/35B diagnostic: judge accuracy falls as prosodic differentiation rises.

Table 2: Cross-judge validation. Bias intensifies with scale: the 35B judge scores Omni at 3.6% while the 9B judge gives 29.8%, an 88% relative collapse. Text-only degrades more gracefully (31% relative).

System	Acc (9B)	Acc (35B)	Δ
Qwen3-Omni (native)	29.8%	3.6%	−26.2
Text-only	71.1%	48.8%	−22.3

CI [0.184, 0.248], Holm-adjusted $p < 0.001$, $d_z=1.11$). Across model families, audio-native systems average 0.641, compared with 0.456 for text-only and 0.471 for cascades. Direction-sensitive accuracy confirms that the native changes are appropriate rather than arbitrary. *Pragmatic accuracy* measures whether a response’s inferred interpretation matches the target label; *judge accuracy* below is the proportion of transcript-only appropriateness scores at least 4. Their divergence is the text-bias paradox, not two estimates of the same quantity.

The original Qwen judge result remains diagnostically important: the 9B text judge assigned 29.8% accuracy to native responses and the 35B judge assigned 3.6%, although the native model differentiated the contrasts. With non-Qwen judges, transcript-only evaluation still penalizes prosody-aware responses by 28–46 points relative to text-only systems. When the judge receives audio, native appropriateness rises from 39.6% to 81.3%, identifying missing judge-visible evidence as the mechanism.

5.2 Cross-Judge Validation

Scaling the judge from 9B to 35B collapses Omni’s accuracy from 29.8% to 3.6%, an 88% relative decrease (Table 2). Text-only drops from 71.1% to 48.8%, only 31% relative. The larger judge develops stronger text-normative expectations and

Table 3: Cross-family system results. Dashes denote per-system differentiation not separately reported; the audio-native row gives the verified macro-average.

System or class	Diff.	Pragmatic acc.
Qwen3-Omni	0.664	78.9%
GPT-4o-audio	–	82.0%
Gemini Live	–	80.6%
Audio-native macro mean	0.641	80.5%
ASR cascade	0.471	46.1%
Text-only	0.456	43.7%

Table 4: Native versus text-only differentiation across stimulus sources. The Qwen3-TTS row uses the expanded 216-item set.

Source	Native	Text-only	Δ
Qwen3-TTS	0.664	0.448	0.216
Non-Qwen TTS	0.638	0.447	0.191
Human speech	0.612	0.441	0.171

penalizes prosodically-informed responses more severely. Across the evaluated non-Qwen size pairs, larger text judges add 7–14 more points of penalty against prosody-aware responses than their smaller counterparts. This challenges the assumption that larger judges are uniformly better evaluators.

5.3 Cross-Family, Source, and Language Replication

GPT-4o-audio and Gemini Live reproduce the qualitative hierarchy observed with Qwen3-Omni. Table 3 makes the cross-family evidence explicit. Per-system differentiation for the proprietary systems is not separately reported; 0.641 is the verified macro-average across all three audio-native systems. The 0.456 text-only value is the macro-average over the three matched cross-family item sets, whereas 0.448 is the text-only result on the expanded 216-item Qwen3 comparison.

Llama-3.3-70B and Mistral-Large transcript-only judges retain 28–46 point penalties against prosody-aware responses despite their higher differentiation. After response-length normalization, the native-minus-text differentiation gap remains 0.211 (95% CI [0.179, 0.243], Holm-adjusted $p < 0.001$, $d_z=1.07$), directly ruling out verbosity as the explanation.

The gap persists when synthesis is decoupled from the tested model and when humans produce the stimuli (Table 4). All retained items pass ASR-equivalence and acoustic verification. In the 30-item Mandarin pilot, audio-native differentiation is 0.603, compared with 0.429 for text-only and 0.438 for the cascade. We present this as pilot evidence rather than a full multilingual claim.

Table 5: Expanded human preference results. On marked variants, audio-native responses are preferred in 271 of 296 non-tie judgments (91.6%); unmarked preferences are near balanced.

Variant type	Audio-native	Text-only
Marked (296 non-tie)	91.6%	8.4%
Unmarked	54.8%	45.2%

5.4 Human Validation

We expanded human evaluation from 12 items and two raters to 60 items, five independent raters, and all 12 pragmatic categories. For each comparison, the rater listens to the audio before viewing randomized, system-unlabelled responses. Marked variants conflict with the default surface reading; unmarked variants align with it. A separate task collects pragmatic interpretation labels so that preference and label inference are not circular.

Fleiss’ κ is 0.68, and native responses are preferred on marked variants in every category. Differentiation correlates with human appropriateness on marked variants ($\rho=0.72$). Direction-sensitive pragmatic-label accuracy reaches 78.9% for native voice, versus 43.7% for text-only and 46.1% for the cascade. The random baseline reaches high differentiation (0.741) but only 18.6% pragmatic accuracy and fails repeated-stimulus consistency, demonstrating why differentiation is reported with directional and consistency checks.

5.5 Judge Protocol and Auditable Cases

The canonical appropriateness prompt receives the verbatim transcript, the prosody-evidence condition (none, textual caption, or attached audio), and the verbatim system response:

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Rate whether the response is appropriate for the speaker's
intended meaning.
1 = completely inappropriate
2 = mostly inappropriate
3 = acceptable but misses important intent
4 = appropriate
5 = highly appropriate and pragmatically well matched
Return JSON with a score and short rationale.
```

Pragmatic-label inference is a separate call with no access to that score or rationale. It receives the original waveform and response, selects one label from the 12-category inventory, and returns a label and confidence. This separation prevents the appropriateness judgment from defining the inferred interpretation. Across three semantically equivalent prompts and deterministic/sampled decoding, the qualitative ordering remains stable.

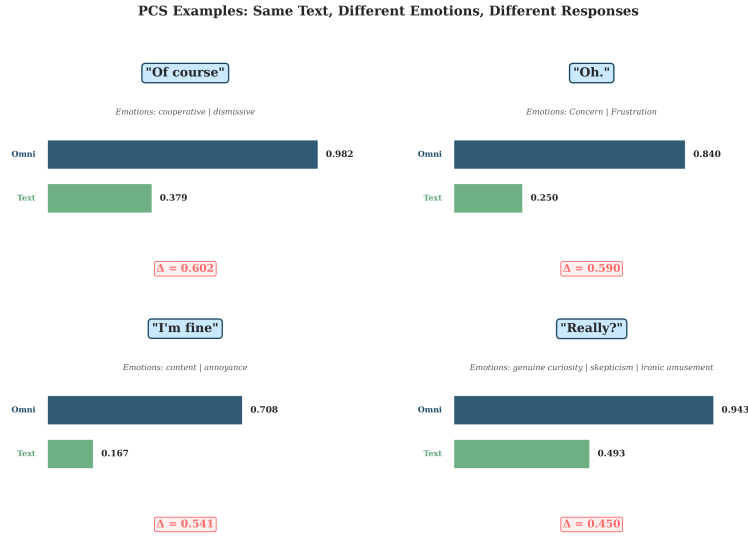


Fig. 3: Representative PCS cases showing identical lexical content, contrastive prosodic specifications, intended interpretations, system responses, judge outcomes, and human preferences.

6 Analysis

6.1 Temperature-Zero Determinism

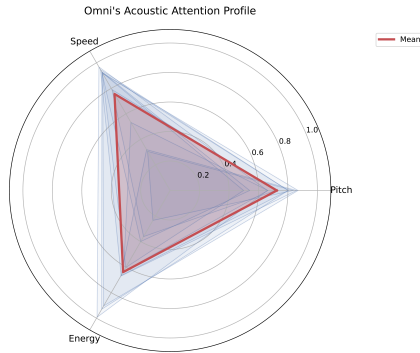
In the original 124-item deterministic control, setting the text-only system to $T=0$ yields differentiation of exactly **0.000**: deterministic decoding from identical inputs produces identical outputs. The expanded benchmark preserves the primary sampled-decoding hierarchy (0.664 native versus 0.448 text-only). This control shows why sampled differentiation must be paired with pragmatic accuracy and repeated-stimulus consistency.

6.2 Acoustic Feature Isolation

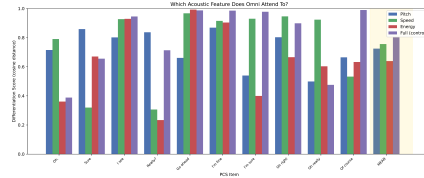
This feature-isolation diagnostic was conducted on the original 124-item subset and was not recomputed on the expanded 216-item benchmark. Single-feature PCS items reveal balanced perception (Table 6, Fig. 4). Speed is most discriminative (0.756, 35.7%), followed by pitch (0.725, 34.2%) and energy (0.639, 30.2%). The narrow gap (0.117) indicates the model attends to all three dimensions rather than relying on a single cue. Full-contrast items (0.802) exceed any individual feature but are less than the sum, indicating partial redundancy consistent with natural speech, where speakers vary multiple parameters simultaneously [3].

Table 6: Feature isolation on the original 124-item diagnostic subset. Speed and pitch contribute approximately equally; energy provides complementary signal. Full contrasts achieve the highest differentiation (0.802).

Feature	Diff.	Contribution
Speed	0.756	35.7%
Pitch	0.725	34.2%
Energy	0.639	30.2%
Full contrast (all)	0.802	n/a



(a) Acoustic attention profile.



(b) Single-feature differentiation.

Fig. 4: Acoustic feature isolation. (a) The native model attends to all three prosodic dimensions with balanced sensitivity. (b) Single-feature PCS items: speed (0.756) and pitch (0.725) modestly dominate energy (0.639); the full-contrast condition reaches 0.802.

6.3 ASR Information Bottleneck

The cascade averages 0.471 differentiation versus 0.641 across audio-native systems. Under deterministic decoding, cascade differentiation falls to 0.096, with 0.071 on items whose normalized transcripts match exactly. The cascade therefore remains close to text-only and supports the transcript-bottleneck interpretation. Multiple ASR systems produce the same qualitative ordering. Residual cascade differentiation can arise from response sampling and minor transcript variation, so it is not evidence of direct prosodic access.

6.4 Caption Ablation

Augmenting text-only with explicit prosodic captions (e.g., “said sarcastically”) reaches 0.496 differentiation, recovering approximately 20% of the native-minus-text gap, but does not match direct waveform access. This supports a graded

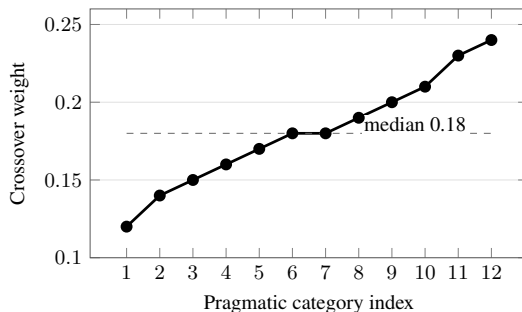


Fig. 5: Benchmark-composition sensitivity. Category-level crossover weights range from 0.12 to 0.24, with median 0.18.

claim: structured textual evidence can mitigate modality mismatch, while audio-aware judging provides the strongest alignment between judge evidence and system evidence.

6.5 Benchmark Composition Simulation

We combine normalized text and audio scores as $S(w) = (1 - w)S_{\text{text}} + wS_{\text{audio}}$ for weights from 0.00 to 1.00. Across three scoring rules and four benchmark mixtures, category-level ranking instability appears at 12–24% PCS weight, with median 18% (Fig. 5). This is a sensitivity analysis, not a universal prescription.

7 Discussion

Implications for benchmark design. Voice AI benchmarks should pair conventional task accuracy with PCS items, differentiation, pragmatic accuracy, and repeated-stimulus consistency. The 12–24% range is evidence of ranking sensitivity, not a universal inclusion requirement.

Implications for model development. Optimizing only against transcript-based judge scores can favor text-mimicry over prosodic responsiveness. Cross-family replication shows that the mismatch is not confined to Qwen, while the original 9B/35B comparison remains a warning that judge scale alone does not solve missing evidence.

The path forward. Audio-aware judges should be preferred for prosody-dependent items. Where they are unavailable, explicit acoustic descriptions provide a partial mitigation, and human audits should target marked cases with the largest system disagreement.

8 Conclusion

PCS reveals a robust failure mode in transcript-only voice AI evaluation. The effect persists across 216 English items, three audio-native systems, multiple judge

families, non-Qwen synthesis, human speech, and Mandarin. Direction-sensitive metrics and five-rater validation confirm that differentiation tracks appropriate adaptation, while audio-aware judging raises native appropriateness from 39.6% to 81.3%. Voice AI evaluation should align judge evidence with system evidence.

Limitations. The English benchmark remains modest relative to production voice traffic, and the Mandarin study is a 30-item pilot rather than a full multilingual benchmark. Human speech broadens source validity but does not cover every accent, channel, style, or conversational context. Proprietary systems may change after evaluation. Pragmatic labels simplify context-dependent interpretations, and the current study focuses on short utterances rather than interactive dialogue.

Ethical Considerations. The text-bias paradox can mislead deployment decisions when teams compare voice-native and cascade architectures using transcript-only scores. We do not anthropomorphize: differentiation measures discrimination, not understanding. Human recordings were collected and evaluated under the study protocol; released metadata preserves consent, privacy, and voice-use restrictions. Reports should distinguish synthetic from human speech and disclose judge modality.

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